Chapter – 5: Changes Around Us: Physical and Chemical

- Few students – discussing changes –
    - Boy – placed cube of ice – half hour ago – converted to water
    - Girl – bud – saw yesterday – rose plant – flower today
    - Girl – bottle – cold water – not cold anymore
    - Boy – yesterday – saw brown spots – banana – today – more spots – strong smell as well
- Activity –
    - You – **obs.** – various changes – something changing – every case
    - Take a moment – **reflect** – changes – each case – **record** obs.
- These changes – size, chape, smell, other prop. – think – other changes – surroundings
- We obs. – changes – around us – help of senses – sight, smell, touch, hearing, state

A Substance May Change in Appearance but Remain the Same

- Activity –
    - Origami –
        - Take – few sheets – paper – fold – create new obj.
        - Unfold – obj. – same paper?
    - Balloon –
        - Take – balloon – inflate – loosen grip – air escapes – uninflated balloon – got back?
        - Take – another balloon – inflate – tighten the opening – prick it – pen – uninflated balloon – got back?
    - Chalk –
        - Crush – small piece – chalk – powder
        - Chalk piece – back from powder – got back?
- All these changes – materials – paper, rubber balloon, chalk – remained same – appearance did change
- Recall – class 6 *curiosity* – water – exist – diff. state – solid, liquid, gas – change state
- All these cases – abs. – change in appearance – no new substance
- Such changes – only phy. prop. – shape, size, state – changes – **physical changes**

A Substance May Change in Appearance and Not Remain the Same

- Activity –
    - Take – 3 glass tumblers – mark them – A, B
    - Glass A – fill 1/4th – tap water – Glass B – fill 1/4th – lime water
    - Blow air – each glass – separate straws – obs. them
- Glass A – blowing air – water – creates bubbles – no change in appearance
- Glass B – blowing air – lime water – creates bubbles, turns milky – leave it – some time – white substance – settles down – indicates – something formed
- Such changes – one or more new substances formed – **chemical changes**
- New substances – formed – process – **chemical reaction**
- This case – CO_2 – exhale – reacts with – lime water – forms new substance – white colour substance – $CaCO_3$ – insoluble in water – liquid appears milky
- Chem. react. – this chem. change – represented – short form – chem. eq.
    - $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

- Lime water – turns milky – test for CO_2
- Activity –
  - Take – tablespoonful – vinegar / lemon juice – test tube
  - Add – pinch – baking soda ($NaHCO_3$)
  - Obs. – fizzing bubbling sound – gas bubbles – formed
  - This gas – pass through – fresh lime water – another test tube
  - Obs. – lime water – turns milky – indicates – gas formed – CO_2
- New substance – CO_2 – chemical change – represented as –
  - Vinegar + Baking Soda → Carbon Dioxide + Other Substances

Some Other Processes Involving Chemical Changes

Rusting
- Iron – rusts – studied – chapter – metals and non-metals – new brown substance formed – rust
- Rusting – chem. change – new substance – iron oxide

Combustion
- Recall – burning – magnesium ribbon – chapter – metals and non-metals – **predict** – phy. / chem. change
- Mg ribbon – burnt – new substance formed – MgO – chem. change
- Obs. – heat, light – also produced – along with – new substance
  - $2\ Mg + O_2 \rightarrow 2\ MgO$ + heat + light
- Chem. react. – substance – reacts with oxygen – produces – heat, light – **combustion**
- Substances – undergo combustion – **combustible** – wood, paper, kerosene
- Activity –
  - Place – 2 candles – separate petri dishes – light them
  - Cover – 1 candle – glass tumbler – obs.
- Open candle – keeps burning – covered candle – stops burning – after some time
- Covered candle – no continuous supply of air – flame – extinguishes
- Component of air – supports combustion – oxygen
- Petri dish – covered candle – add lime water – turns milky
- CO_2 – carbon from wax – oxygen from air

Is presence of air enough for combustion?
- Learnt above – combustible substances, oxygen – required – combustion
- Paper – combustible – keep in air – any amount of time – does not burn
- Activity –
  - Hold – piece of paper – tongs – light it – match stick
  - Take – another piece of paper – use – magnifying glass – focus – sunrays – smallest, brightest spot – hold there – some time
- Obs. – paper – emit smoke – catches fire – substance – burn without fire
- Focusing sunrays – paper heats up – temp. increases
- After some time – paper – so hot – catches fire
- This min. temp. – substance catches fire – **ignition temperature**
- Temp. – lighted matchstick – already higher – ignition temp. of paper – catches fire immediately
- **Conclude** – 3 req. – combustion –

- Combustible substance – fuel
- Oxygen
- Heat – allows – fuel – reach – ignition temp.

Can Physical and Chemical Changes Occur in Same Process?

- Activity –
  - Burning candle – students obs. – analyse
  - Wax – melts, evaporates
  - Wax – melts, flows, solidifies – diff. shape
  - Wax – burns – new substances – formed
  - Wick – draws up the liquid wax
- Wax – candle – melts – carried up – wick – evaporates – heat of flame
- Melting, solidification, evaporation – wax – phy. changes
- Burning of vapour – chem. change
- Burning of candle – both – phy., chem. change

Are Changes Permanent?

- Once changed – can we regain – original form
- Activity –
  - Think again – all the changes – discussed so far
  - Which changes – regain – original form – record obs.
- Return to original obj. – started with – changes – **reversed**
- Ex. – ice melts – refrozen to ice
- Water evaporates – condensed back – liquid water
- Some changes – cannot reverse – original obj. – cannot be regained
- Ex. – chopped vegies – cannot return – original form, shape
- Changes around us – grouped – reversible, irreversible

Are All Changes Desirable?

- Many changes – daily life – useful
- Ex. – milk to curd – ripening, cutting, cooking of foods
- All these – **desirable** changes – some changes – **undesirable**
- Ex. – rusting of iron – decay of food – storage
- Change – undesirable – some situations – desirable – other situations
- Ex. – decomposition of food – very useful – producing compost
- Some changes – over years – human activity – long-term environmental impact
- Ex. – increased consumption – fuels – cars, trains, aeroplanes – increasing amt. – CO_2
- Drying of paint – walls, doors, furniture – releases many substances – causing atmospheric pollution

Some Slow Natural Changes

Weathering of rocks

- Heaps – sand, soil, stones – base of mountains – sediments

- Formed – phy. changes – break up – large rocks – smaller pieces
- Temp. changes – climate conditions – growing roots – freezing of water within cracks – rocks break
- Water / chem. in water – react with rocks – cause chem. changes – their composition
- Ex. – originally black basalt – contain iron – changes chemically – red layer
- Red colour – result – iron oxide – rock surface – exposed – long time – water / air containing water vapour
- These phy., chem. changes – collectively – **weathering** – leads – formation – soil

Erosion

- Notice – fine sand – riverbeds, lakes
- This sand – formed – rock, pebbles, soil, sediments – break down – move – one location to another – natural forces – wind, flowing water
- This process – **erosion**
- Erosion – ex. – phy. change
- River rocks, pebbles – appear smoother – constant erosion – flowing water
- Speed of water, wind – decreases – ocean, lake – materials transported – settles down
- These sediments – harden over time – become rocks
- Most of these changes – over 1000s of years – cannot be reversed